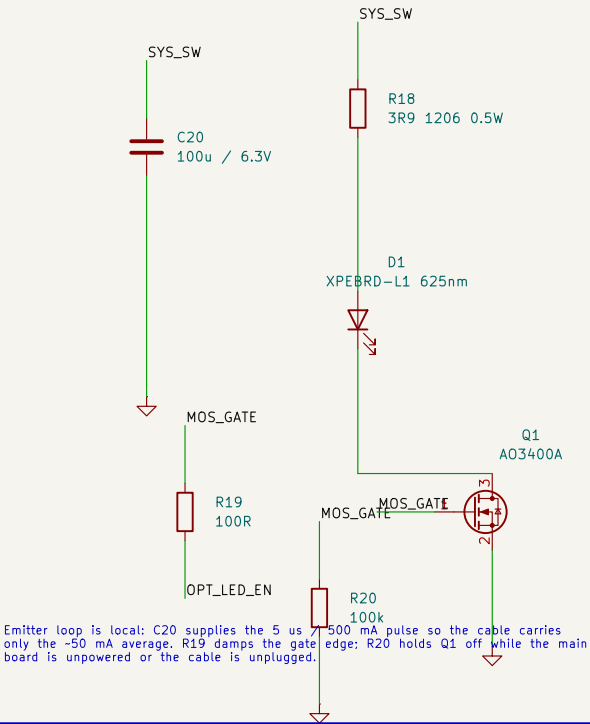
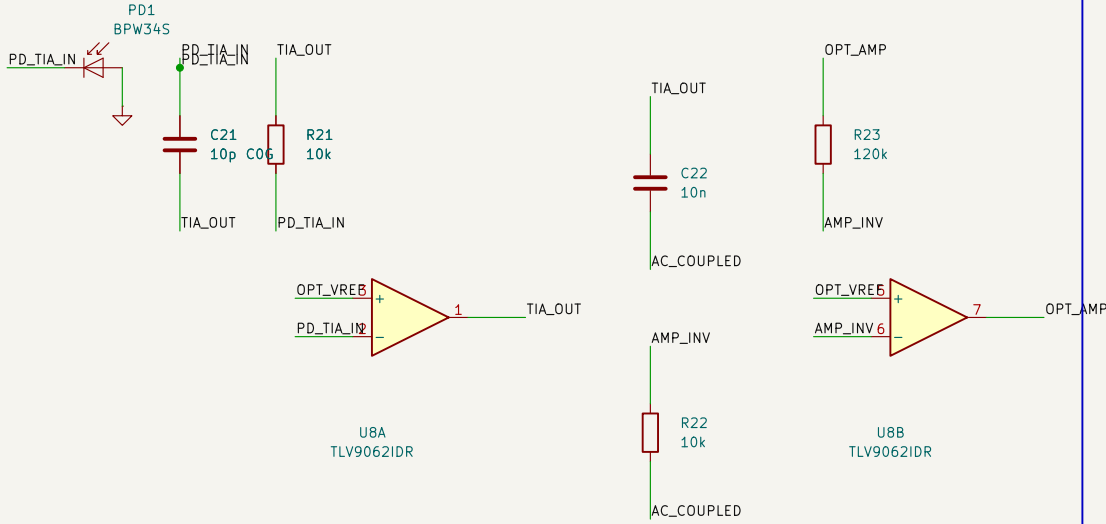


PULSED EMITTER – 20 kHz / 5 us / ~500 mA

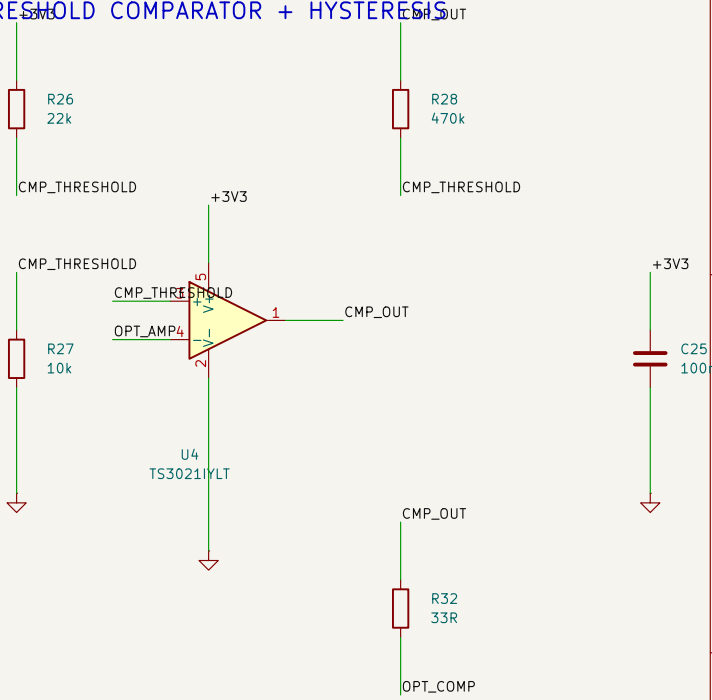


Emitter loop is local: C20 supplies the 5 us / 500 mA pulse so the cable carries only the ~50 mA average. R19 damps the gate edge; R20 holds Q1 off while the main board is unpowered or the cable is unplugged.

TRANSIMPEDANCE + x12 AC GAIN

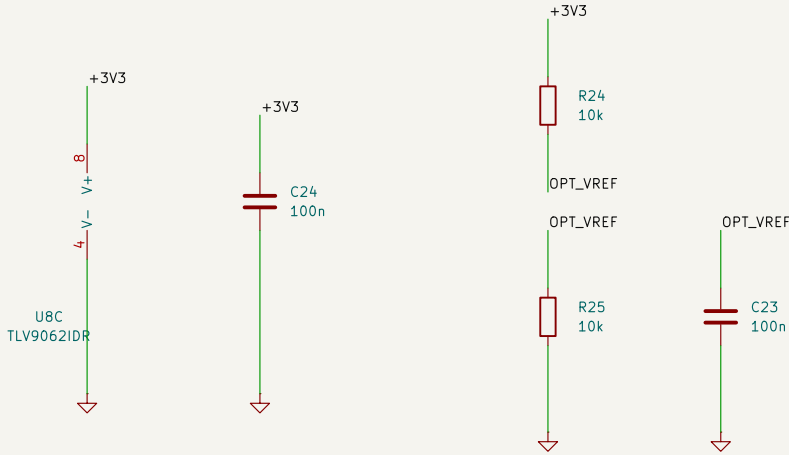


THRESHOLD COMPARATOR + HYSTERESIS

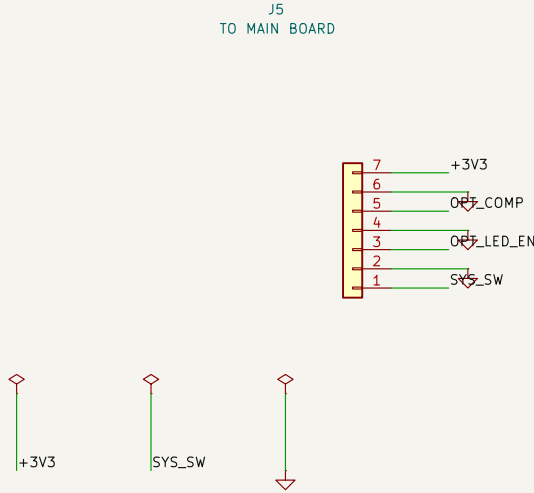


Ground sits between OPT_LED_EN and OPT_COMP. Emitter-pulse crosstalk onto OPT_COMP lands inside the synchronous gating window and would be accepted as a genuine hit, so it must be stopped here, not filtered downstream. Use a latching, gold-plated JST series with secondary retention: tin friction contacts fret under sustained vibration and fail intermittently rather than cleanly.

1.65 V REFERENCE



CABLE TO MAIN BOARD



Ground sits between OPT_LED_EN and OPT_COMP. Emitter-pulse crosstalk onto OPT_COMP lands inside the synchronous gating window and would be accepted as a genuine hit, so it must be stopped here, not filtered downstream. Use a latching, gold-plated JST series with secondary retention: tin friction contacts fret under sustained vibration and fail intermittently rather than cleanly.

BalancerREF_OptHead – optical tach head, at 90 degrees to the main PCB on the aluminium L-bracket. One pivot bolt aims the emitter; add a second fastener (arc slot or serrated lock nut) or it drifts under vibration and loses RPM lock. The baffle between D1 and PD1 and the red acrylic window belong to this board. Designators match BalancerREF Rev C so the existing pin audits stay valid.

Cable carries DC rails and digital edges only
Split from BalancerREF Rev C – human review required

Sheet: /
File: BalancerREF_OptHead.kicad_sch

Title: BalancerREF_OptHead – pivoting optical tach head

Size: A3 Date: 2026-09-11 Rev: B
KiCad E.D.A. 10.0.3 Id: 1/1